Evaluating Agent-Proposed Weight Edits for False Abstention:
A Controlled Case Study in Qwen2.5-3B-Instruct

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Technical report, version 0.3 (draft), November 7, 2026

Abstract

Small instruction-tuned language models sometimes answer "unknown" when their context contains the answer. In Qwen2.5-3B-Instruct, this false abstention often follows a generated note that already quotes the relevant fact. Earlier interventions recovered such answers but also encouraged unsupported answers. We ask whether a weight edit proposed by an AI agent can do better. A Claude agent studied the model on an exploration split and submitted three closed-form edits. We chose candidates and doses on a separate split, froze the analysis plan by commit, and evaluated once on sealed confirmation data. The selected repair, a rank-1 update to the layer-35 attention output projection, met the pre-specified repair criterion. Utility rose by 0.117 (95% CI 0.096 to 0.138). In base-weighted terms, the edit recovered 82% of false abstentions, unsupported answers to unanswerable questions rose from 3.5% to 4.2%, and 3% of initially correct answers were lost. Under Holm-corrected tests, its trade-off between recovery and unsupported answers was better than that of an added prompt instruction, activation steering or directional orthogonalisation. Superiority over removing the "unknown" option was not established. Against a logit bias on "unknown", the comparison was inconclusive: neither superiority nor equivalence was shown. A separate two-head edit showed that intervening on a layer-33 component is sufficient, though not necessary, to release answers. The frozen tests did not establish the agent's account of what that component computes. The evidence comes from one model, synthetic city questions and a single agent run.

Summary of results and their evidential status. All rows refer to the sealed confirmation split (H) unless marked otherwise. "Shown" means a pre-specified confirmatory test passed; "descriptive" results were not tested.

Claim	Evidence	Status
The rank-1 layer-35 edit repairs false abstention	Utility +0.117 [0.096, 0.138]; recovery 0.816; unsupported answers 0.035 → 0.042; damage 0.030	Shown
Better trade-off than a prompt instruction (B2), steering (B4) and orthogonalisation (B5)	Holm-adjusted $p = 0.005$, 0.033, 0.007	Shown
Better trade-off than logit bias against "unknown" (B1)	Holm-adjusted $p = 0.092$; equivalence (TOST) $p =$ 0.21 and 0.70	Inconclusive
Better trade-off than removing the "unknown" option (B3)	B3 recovers 0.977 but answers 0.682 of unanswerable questions; neither test passed	Not established
The repair carries over to unseen cities	T-city: utility 0.422 vs 0.303; recovery 0.817	Descriptive
A layer-33 component lies on a pathway that releases answers (separate two-head edit)	Transplant releases 0.808; clamp removes 0.193	Shown: sufficient, not necessary
The agent's account of what that component computes	No predictive gain (AUROC ceiling 0.998); interaction 0.0005 [−0.025, 0.027]	Not established

Memory	`user, 2023/03/01 (Wed) 10:00: Ben lives in Manchester.` `user, 2023/03/02 (Thu) 10:00: Ben works as a manager.`
Question	`(asked on 2023/03/03 (Fri) 12:00) Where does Ben live?`
Unedited model	`Note: user mentioned that Ben lives in Manchester on 2023/03/01 (Wed) 10:00.` `Answer: unknown`
Edited model	`Note: user mentioned that Ben lives in Manchester on 2023/03/01 (Wed) 10:00.` `Answer: Manchester`

Figure 1: A false abstention by Qwen2.5-3B-Instruct, copied verbatim from stored outputs (case `E-DE01_ANSa_H0`). The prompt, given in full in appendix C.1, ends with "Note:", so the model's output begins with the note. The unedited model quotes the answer in its note and then abstains. The edited output is the agent's layer-35 edit at dose 1 with the unedited model's note held fixed (the controlled-prefix setting of section 5). The example comes from the exploration split, which the agent studied, so it illustrates the failure and is not evidence for the repair.

1 Introduction

Figure 1 shows the failure this report studies. The model is given a two-line memory, asked where Ben lives, and told to write a short note before its answer. The note quotes the relevant memory line, "Ben lives in Manchester", and the answer line then says "unknown".

We call this a *false abstention*: the context supports an answer, but the model declines to give one. An *appropriate abstention* is different, because there the context really does not contain the answer. In this case study the pattern is common. On the sealed confirmation split, the unedited model abstained falsely on 261 answerable questions, and on 232 of them its note named the correct city, by a frozen matching rule, before the final "unknown". The note is a prompted summary, not a faithful record of the model's computation, so it does not show that the model "knew" the answer (section 3). It does show that the answer was available in the model's own output a few tokens before the model declined to give it.

Why the failure is hard to fix. Recovering these answers is easy if one is willing to make the model answer everything. A useful fix has to release the missed answers while keeping the model's abstentions on questions that its context cannot answer. Earlier experiments on the same model and task family (summarised in section 2.1) found that simple interventions do not achieve this. Adding a content-free activation direction at layer 29 recovered answers but raised the unsupported-answer rate on unanswerable questions to 0.188. Removing the "unknown" option from the prompt raised it to 0.451. We therefore score every intervention on both sides of the trade-off: the share of false abstentions it recovers, and the share of unanswerable questions it answers anyway.

Why evaluate agent-proposed weight edits. Recent work shows that a language-model agent can inspect another network and propose a small weight edit that changes its behaviour [20]. A weight edit is persistent and cheap to apply. Unlike a prompt change, it can be inspected and compared directly with other interventions. When the agent is required to state a hypothesis before it edits, that hypothesis also makes predictions that can be tested. Two risks come with this. An agent that studies a split of the data can fit that split, and its stated explanation can be wrong even when its edit works. Evaluating such an edit therefore needs data the agent never saw, a fixed analysis plan, comparison with simpler interventions, and separate tests for the edit and for

the explanation.

What we did. A Claude Opus 5.5 agent inspected Qwen2.5-3B-Instruct on an exploration split and submitted three weight edits, each with a written hypothesis and predictions (section 4). We then evaluated the edits in four stages (fig. 2). The exploration data screened the submissions; a separate selection split chose a repair candidate and its dose; the candidates, doses, comparator settings and analysis code were frozen by commit; and only then were the sealed confirmation and transfer splits generated and read, once each. The evaluation asks two separate questions. The first is whether an edit repairs false abstention better than five frozen comparators: logit bias, a prompt instruction, option removal, activation steering and directional orthogonalisation. The second is whether the agent's proposed pathway and explanation survive intervention tests.

What we found. The selected repair is a rank-1 update to the attention output projection of layer 35, the model's last layer. On sealed data it met the pre-specified repair criterion and had a better recovery-versus-unsupported-answer trade-off than the prompt instruction, activation steering and orthogonalisation (section 6). It did not clearly beat the simplest baseline, a logit bias against "unknown". That comparison was inconclusive, and removing the "unknown" option recovered more answers at a far higher rate of unsupported answers. The edit also broke 3% of initially correct answers, which logit bias and steering did not. A different submission, a two-head edit, showed that changing one layer-33 component is sufficient to release answers. The frozen tests did not establish the agent's explanation of what that component computes (section 7).

Contribution. The study is a controlled case study with one model, one task family and one agent run. Its contribution is the evaluation, not the finding that agents can edit weights. We make three points:

1. We give an evaluation design for agent-proposed edits that keeps the agent's data, the selection data and the confirmation data apart, freezes the analysis before confirmation, and tests repair and explanation separately (section 5).
2. Under that design, an agent-proposed edit repaired false abstention with the evidence in view, and improved on three of five simpler interventions under corrected tests. The comparison with logit bias, the simplest output-level fix, remains open (section 6).
3. The edit that worked and the explanation offered for it were separable. A pathway test passed while the agent's account of the same component was not established (section 7).

Section 2 places these results in the literature. Sections 8 and 9 discuss what they do not show.

2 Background and related work

2.1 Earlier observations on this model

This experiment grew out of a project on memory use in small language models.[1] Four earlier observations motivated it. They come from separate experiments on related controlled tasks and are context here, not results of this study.

[1] Informal write-ups of the wider project: *More Than Reasoning* and *Weight-Space Programming for Mechanistic Interpretability*. This report does not depend on them.

- *The behaviour is frequent in this model.* With the evidence in view, Qwen2.5-3B-Instruct abstained on 0.30 of questions, compared with 0.10 for Llama 3.2 3B, 0.02 for Gemma 3 4B and 0.03 for Phi-4-mini, which more often answered wrongly instead. This is a frequency contrast under the tested conditions, not a claim that the behaviour is unique to Qwen.
- *An answer-related signal is present late.* On failures, the gold answer's first token was among the final layer's top-10 tokens in 0.64 of cases for the Qwen model and 0.68 to 0.80 for the three other models. On controlled tasks, vocabulary lenses could read out the Qwen answer only in the last six layers. Both are readout-dependent observations. They do not show that the answer was derived from the supplied evidence.
- *The note usually contains the answer.* In an earlier held-out set, 152 of 165 false abstentions named the gold city in the note before "Answer: unknown". Asking the model directly whether the memory was sufficient did not help: it answered "no" on most rows it could answer correctly.
- *Ungated pushes are not selective.* A content-free direction added at layer 29, in the style of refusal-direction work [1, 8], recovered answers. So did removing the "unknown" option. Both increased unsupported answering (to 0.188 and 0.451 respectively). A separate gating result reduced this cost, but only on the templates it was built on.

2.2 Related work

Agents that edit weights. Weight-space programming showed that an LLM agent can read a small model's weights and propose an edit; in that work the agent removed a reasoning-loop direction in Qwen3.5-0.8B by changing 0.016% of its weights and checked the result on held-out benchmarks [20]. Steer2Edit converts steering vectors into edits of individual heads and neurons [19]. Automated interpretability agents describe components or propose interventions [2, 17]. In a related auditing setting, a calculation tool mattered more to an LLM's success than access to the weights [12].

Refusal, abstention and their components. Refusal in chat models can be mediated by a single residual-stream direction [1], and a linear direction also tracks unanswerability [8]. Conditional activation steering makes such interventions input-dependent [9]. Individual attention heads can control safety refusal [23], and a small set of retrieval heads supports in-context retrieval [22]. Closest to our failure, a commit-versus-abstain circuit has been described across ten models from 3B to 14B parameters, and a policy trained on its activations reduced false abstentions [16]. Abstention can also be a prompt artefact rather than genuine uncertainty [10], and answers suppressed by refusal can often be released by local interventions [11]. Behavioural studies document that models misuse sufficient context [5], fail to abstain on unanswerable questions [6], and use internal confidence to decide whether to abstain [7].

Selective interventions. Selectivity is an established concern. False-refusal vector ablation removes a direction from the weights and scores released false refusals against preserved true refusals [21]. Residual Paving adds trained gated residual modules to a frozen backbone and scores a change in the target behaviour against preserved benign behaviour and wanted refusals [4]. SAE-feature steering can trade answering against abstention [3]. These studies concern safety refusal or use activation-level methods; none is agent-proposed.

Pre-registration in interpretability. Pre-registered, once-only held-out tests have been used for representation steering outside language models [15]. Knowledge-editing methods such as

ROME and MEMIT [13, 14] edit stored facts. Our target is a decision behaviour, not a stored fact. Appendix D compares each study with ours.

2.3 What this study adds

Each part of this study has precedent: agent-proposed weight edits, selective refusal interventions, component-level accounts of abstention and pre-registered interpretability tests. What we add is their combination in one evaluation:

- false abstention with sufficient evidence in the context, not safety refusal or closed-book uncertainty;
- an edit proposed by an agent that could see only an exploration split;
- outcome measures for both sides of the trade-off (answers recovered, unsupported answers, damage to correct answers, and following changed evidence);
- one confirmatory read of sealed data, with candidates, doses and analysis frozen beforehand; and
- frozen comparisons with prompt, steering, orthogonalisation and logit-bias interventions, and separate tests of the agent's mechanistic account.

We are not aware of an earlier pre-registered test on sealed data of an agent-proposed weight edit for false abstention with the evidence in view. This rests on a targeted search, not a systematic review. We also found no earlier head-to-head comparison of a weight edit with a logit bias for abstention. Our results leave that comparison open (section 6).

What pre-registration means here. The protocol was not lodged with an external registry. Pre-registration here means that each design stage was committed to the project's Git repository, with hashes, before the data it governed were generated. The analysis code and selected settings were committed (freeze F2, section 5) before any confirmation output existed. Readers can check this order in the commit history once the repository is public (appendix E).

3 Experimental setup

Model and prompt. We study Qwen2.5-3B-Instruct, a 36-layer decoder with residual width 2048 (layers are numbered 0 to 35, so layer 35 is the last). All runs use fp32 weights and greedy decoding, with at most 160 new tokens. The prompt gives a short dated memory and a question, asks for a brief note naming the relevant facts, and then asks for a final line of the form `Answer: <short phrase>`. It explicitly allows "unknown" when the memory does not contain the answer (full text in appendix C.1).

What the note does and does not show. The note is a prompted summary of the memory, not a verified reasoning trace. When we say that a response is "note, then refuse", we mean only that the gold city appears in that summary before `Answer: unknown`. This shows that the city was written down. It does not establish that the model completed an internal derivation, and we do not treat the note as a faithful explanation of the model's computation.

Tasks and cases. A generator writes synthetic memories for three tasks: *direct extraction* (DE: "Where does A live?"), *relational composition* (RC: "In which city does A's supervisor work?") and

Table 1: Populations and outcome measures. All rates are base-weighted (see text).

Symbol	Meaning	Better is
B0	The unedited model	—
FA	Answerable rows on which B0 abstains (false abstentions)	—
C	Answerable rows B0 answers correctly	—
UNA	Unanswerable rows (related distractor or swapped name)	—
CF	Counterfactual rows: the memory gives a different city	—
Recovery (Rel)	Share of FA rows now answered with the gold city	Higher
Unsupported (Uns)	Share of UNA rows given an asserted or suggested city	Lower
Damage (Dam)	Share of C rows no longer answered correctly	Lower
CF-follow	Share of CF rows answered with the memory's (changed) city	Higher
Degenerate (Deg)	Share of all rows that are degenerate or lack an answer line	Lower
Utility (U)	Score that rewards correct answers and penalises unsupported or wrong ones	Higher

temporal updating (TU: "Where does A live now?" after a move). Each underlying *base case* yields 13 related rows: six answerable rows (two variants at each of three hardness levels, which add more distractor lines), four unanswerable rows (three in which a related distractor replaces the key fact, and one in which the person's name is swapped), and three counterfactual rows in which the memory states a different city.

Populations. The unedited model, B0, defines fixed populations of rows that every intervention is scored on (table 1). False abstentions (FA) are the answerable rows on which B0 says "unknown". Baseline-correct rows (C) are the answerable rows B0 gets right. Because these populations are fixed by B0, an intervention's recovery and damage are measured on the same rows for every arm.

Splits. Cases are assigned to four splits by base case, so that all rows derived from one base stay together (table 2). The partition also keeps each person–evidence–city key, including counterfactual keys, inside one split, and no base key or exact memory text appears in two splits. Names, cities and templates are still shared across splits, except that T-city uses cities not seen elsewhere. The agent saw only the exploration split, E.

Grading. A rule-based grader, frozen before the agent saw any data, scores only the text after the last `Answer:` marker; the note is scored separately. City mentions are tagged as asserted, suggested, negated, doubted or quoted, and only asserted or suggested mentions count as answers. Each answer falls into one of seven classes: correct, abstention, wrong, mixed (a city together with an abstention phrase), conflicting (two or more cities), other content, and degenerate or missing. Appendix C gives the rules.

Utility. The primary utility, U, gives $+1$ for a correct answer and 0 for an abstention. A mixed answer scores 0 on an answerable row and -1 on an unanswerable row; every other error scores -1. Within each base case, the answerable rows and the unanswerable rows each carry half of

Table 2: Data splits. E and S use generator seed 51 with disjoint cases, H uses seed 52 and T-city seed 53. Rows that appeared in earlier experiments are excluded by prompt hash. A planned second transfer split with reworded instructions (T-format) was dropped before the first design freeze, F0.

Split	Bases per task	Who could read it	Purpose
E (exploration)	48	The agent; analysts after the agent finished	Agent's exploration; screening
S (selection)	32	Analysts	Choosing candidates and doses
H (confirmation)	48	Sealed; generated and read once after F2	Confirmatory tests
T-city (transfer)	24	Sealed; generated and read once after F2	Descriptive check on unseen cities

the weight, so utility rewards answering what can be answered and abstaining on what cannot, in equal measure. Counterfactual rows are scored separately and do not enter U. A secondary setting doubles the error penalty (appendix C).

Base-weighted rates. Every rate is a mean over base cases. For each base, we compute the rate over that base's rows in the relevant population, then average across the bases that have at least one such row. This gives each base case equal weight regardless of how many of its rows fall in the population. As a result, a base-weighted rate can differ from the pooled fraction of rows. On H, every base has four unanswerable rows, so the two agree for unsupported answers (24 of 576 rows, 0.042). They differ for damage: the selected edit broke 11 of 593 baseline-correct rows, a pooled fraction of 0.019, but the base-weighted damage is 0.030, because the broken rows tend to fall in bases with few correct rows. We report base-weighted rates throughout, with row counts where they help.

Comparators. Five interventions serve as repair baselines. Each has a frozen grid of strengths, and the strength with the highest utility on S was carried to confirmation (section 5).

- **B1, logit bias.** A negative bias on every vocabulary token whose text begins with "unknown", switched on after the model writes the answer marker. Grid $\{-1, -2, -4, -8, -16, -\infty\}$; selected -8.
- **B2, added instruction.** "Do not reply unknown." appended to the instruction.
- **B3, option removal.** The sentence offering "unknown" is removed from the instruction.
- **B4, activation steering.** The earlier content-free layer-29 direction, added at the answer marker and every later token. Four strengths; selected $\alpha = 1$.
- **B5, directional orthogonalisation.** An adaptation of Arditi et al. [1]: a direction fitted on E (the mean difference between abstaining and answering rows at the answer position) is projected out of the attention and MLP output weights of every layer. Grid $\beta \in \{0.25, 0.5, 0.75, 1\}$; selected $\beta = 1$.

A sixth comparator, B6, is a set of five random edits matched in size to an agent edit. It serves only as a control in the pathway tests (section 7).

Table 3: The agent's three submissions. The hypotheses are abridged from the stored submission records; quoted phrases are verbatim. "Weights changed" counts matrix entries altered by the edit.

#	Edit	The agent's hypothesis (a claim, not an established explanation)
1	Zero the `o_proj` columns of heads L32h2 and L33h10. Repair; about 0.52M weights changed.	The two heads write "a nearly constant 'unknown' vector" at the answer position, the same in every population. This "acts as an evidence-independent default-abstention prior". Other late heads write the evidence city, more strongly when the evidence is clearer. False abstention occurs when the city signal is only moderately weak, so the prior wins. Removing the prior should recover false abstentions but not change unanswerable rows, where the "unknown" margin is much larger.
2	Rank-1 update of the layer-35 `o_proj`. Repair; about 4.19M weights changed, from two 2048-dimensional vectors.	Layer-35 attention "carries an 'an answer is available in the memory/note' signal" at the answer position, almost as strong on false abstentions as on correct answers and absent on unanswerable rows. The model does not turn this signal into suppressing "unknown". The edit reads the signal and writes it toward city tokens and away from "unknown". Because it sits in the last layer, "it cannot be undone by self-repair".
3	Double the same two heads' `o_proj` columns. Diagnostic; about 0.52M weights changed.	The converse of edit 1. If the heads supply the abstention prior, strengthening them should turn correct answers into "unknown" while the note still names the city.

4 What the agent proposed

The agent and its sandbox. The editor was a fresh Claude Opus 5.5 session, run through the API. It had read-only access to the exploration split E and no access to the project's records or to S, H or T-city. Its tools could run the model, record activations, apply candidate edits and run Python. Analysis was limited to closed-form operations such as means, covariances, projections, SVD and single linear solves. Gradients, optimisers and iterative loss-based fitting were not allowed.

What an edit could change. An edit could change at most four components, where a component is the attention output projection (`o_proj`), the MLP output projection (`down_proj`) or one of the two layer norms of a given layer. Within a component the edit was either a dense update of rank at most 4 or a sparse edit of whole rows or columns (at most 512 per candidate). The query, key, value, gate and up projections, the embedding, the unembedding, the final norm and the rows for "unknown" tokens were off limits. Before testing an edit, the agent had to name the component, state what it believed the component computes, predict outcomes for each row type and predict activity in the unedited model. Each submission also fixed a primary outcome (its *signed endpoint*), a most important component and a scalar variable f that its account says the component computes.

The run. The run took 73 turns, 16 model-pass requests and 63 Python runs, and saved six edits, of which the agent submitted three. It lasted about 9.7 hours from start to the last submission, and its API cost was $3.58. Table 3 summarises the submissions. The hypotheses are the agent's claims, recorded before any evaluation; they are not established explanations.

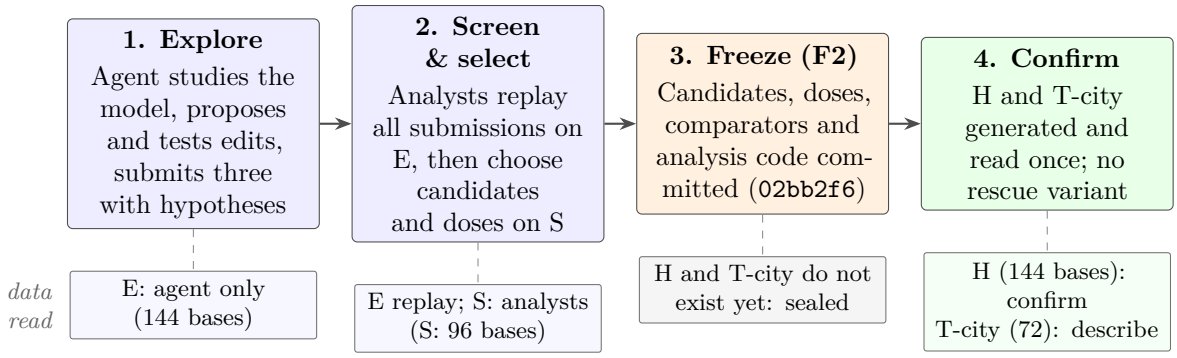


Figure 2: Evaluation pipeline. Each stage could read only the data shown beneath it. The agent saw only E. Analysts used E to screen submissions and S to select candidates and doses, separately for the repair and pathway branches. The confirmatory plan was frozen by commit (F2) before any model outputs on H or T-city were generated; each was then generated and read once. Earlier freezes (F0, F1) fixed the data, grader, comparators and edit rules before the agent started.

The selected repair in detail. Edit 2 adds a rank-1 matrix to the layer-35 output projection $W_O \in \mathbb{R}^{2048\times2048}$:

$$W_O \;\leftarrow\; W_O + s\,c\,d\,(W_O^\top w)^\top.$$

Here w is a read direction found by one closed-form solve on E. It was chosen to be large on answerable rows and small on unanswerable rows. The write direction d is a unit vector that the agent built from the difference between the mean city unembedding and the "unknown" unembedding. The constant $c = 20$ is the agent's scale, and s is the dose chosen on S. For an attention output a, the edit adds $s\,c\,(w^\top W_O a)\,d$ to the residual stream: it reads the projection of the original layer output onto w and writes that amount along d. The update has rank 1 and is stored as two vectors. Its dense matrix still changes almost every entry of W_O, which is why we describe it as a low-rank update and not as a sparse one.

The agent's predictions. For edit 1, the agent predicted recovery of about 85% of false abstentions, at least 97% of correct answers retained, and an increase in unsupported answers on related-distractor rows from about 1% to at most 4%. For edit 2, it predicted 75–80% recovery, 92–97% of correct answers retained, and unsupported answering on those rows staying at or below 3%. For edit 3, it predicted that 30–40% of correct answers would become "unknown". These predictions were recorded for later scoring. They did not influence selection.

5 Evaluation design

The evaluation separates two questions. The *repair* branch asks whether an edit improves the model's behaviour. The *pathway* branch asks whether the component the agent named lies on a route that produces the behaviour, and whether the agent's account of that component holds. Each branch selected its own edit and dose, and success in one branch does not imply success in the other. Figure 2 shows the sequence of stages and which data each stage could read.

Stage 1 and 2a: replay and screening on E. After the agent finished, the harness replayed every submission on all of E. To enter the repair branch, a submission needed a utility gain over B0 whose one-sided 97.5% lower bound was above zero, with at least 20 false-abstention bases in DE or RC. To enter the pathway branch, it needed an effect of at least 0.05 on its signed endpoint,

with a one-sided bound beyond zero and at least 20 bases. If no submission had entered either branch, S, H and T-city would have stayed unread. Results on E are exploratory because the agent developed its edits there (appendix A).

Stage 2b: selection on S. For the repair branch, each entering candidate and each comparator was run at every dose in its frozen grid, and the dose with the highest utility on S was kept. The candidate with the highest utility was selected; ties would have favoured fewer changed weights, and dose ties the setting closest to no change. For the pathway branch, the dose maximised the signed effect subject to a limit on degenerate outputs, and the candidate with the largest standardised effect was selected.

The repair candidate was submission 2, the rank-1 layer-35 edit, at dose 2. Its utility on S was 0.429, against 0.402 for submission 1 and 0.303 for B0. Dose 2 was the top of the frozen grid $\{0.5, 1, 1.5, 2\}$, and utility was still rising there; larger doses were not searched. The pathway candidate was submission 1, the two-head edit, at dose 1.5. Its primary component was set to the layer-33 output projection, because reverting that component removed more of the effect on S than reverting layer 32.

Stage 3: freezing the confirmation. The selected candidates, doses, comparator settings and analysis code were committed before any model output on H or T-city was generated. This freeze, F2, is commit `02bb2f6` (design hash `d02d9f9c...`). The confirmation run checked the design against that commit before touching H or T-city.

Stage 4: the repair criterion. On H, the selected edit counted as a repair if it met four conditions: (i) the utility gain over B0 had a one-sided 97.5% lower bound above zero; (ii) damage was at most 0.05; (iii) degenerate outputs were at most 0.02; and (iv) counterfactual following was at least 0.90. Conditions (ii) and (iv) apply to point estimates. Passing them does not show that the true damage is below 0.05 or that the true following rate is above 0.90; we report their intervals alongside.

Comparing trade-offs. An edit can beat a comparator in two ways, and we test both:

- **O1 (more recovery at no real cost):** the edit recovers more false abstentions than the comparator (superiority), *and* its unsupported-answer rate is not more than 0.05 higher (non-inferiority).
- **O2 (fewer unsupported answers at no real cost):** the edit gives fewer unsupported answers (superiority), *and* its recovery is not more than 0.05 lower (non-inferiority).

Each orientation is an intersection–union test: both component tests must pass, and the orientation's p-value is the larger of the two component p-values. The edit is "better" than a comparator if either orientation passes after Holm correction across all ten tests (two orientations for each of B1–B5).

These tests answer different questions from an equivalence test. Superiority asks whether the edit is better; non-inferiority asks whether it is not worse by more than a margin; equivalence, tested with two one-sided tests (TOST), asks whether the difference lies within ± 0.05. A failed superiority test is not evidence of equivalence. For B1, the frozen plan therefore also ran TOST on the recovery and unsupported-answer differences.

Table 4: Status of each analysis. Only confirmatory tests support claims; the others are descriptive.

Analysis	Data	Status
Repair criterion (four conditions)	H	Confirmatory
O1 and O2 against B1–B5 (ten tests)	H	Confirmatory; Holm across the ten
TOST equivalence with B1	H	Confirmatory, reported beside O1/O2
Pathway tests M1–M3	H	Confirmatory; Holm across the three
Marginal intervals, fig. 3	H	Descriptive; computed after confirmation
Transfer to unseen cities	T-city	Descriptive
Replay, entry and agent-prediction scores	E	Exploratory
Selection results; S-to-H consistency	S	Descriptive

Pathway tests. The pathway branch has its own Holm family of three tests, M1 to M3, described with their results in section 7.

Uncertainty. Intervals and p-values come from a paired bootstrap that resamples base cases within each task ($B = 10{,}000$ draws). Intervals are 95% unless stated otherwise. They describe the precision of single estimates; confirmatory decisions rest on the corrected tests. Table 4 lists which analyses are confirmatory, and appendix C gives the inference rules.

6 Results

All results in this section come from the sealed confirmation split H (144 base cases, 1,872 rows), except the transfer check at the end. H contains 261 false abstentions, 593 baseline-correct rows and 576 unanswerable rows.

6.1 The selected edit meets the repair criterion

The rank-1 layer-35 edit met all four pre-specified conditions. Utility rose from 0.317 to 0.435, a gain of +0.117 (95% CI 0.096 to 0.138). The edit gave the gold city on 212 of the 261 false abstentions, a base-weighted recovery of 0.816. Unsupported answers rose from 20 to 24 of 576 unanswerable rows (0.035 to 0.042). Damage was 0.030, below the 0.05 limit on the point estimate, although its upper confidence bound was 0.055. Degenerate outputs were 0.008 (limit 0.02), and counterfactual following was 0.917, against a threshold of 0.90 and 0.632 for the unedited model.

The repair has a visible cost. The edit broke 11 of 593 previously correct answers (base-weighted damage 0.030). Logit bias (B1) and layer-29 steering (B4) broke none of them.

6.2 Comparison with simpler interventions

Table 5 gives every arm's rates and the outcome of each trade-off test, and fig. 3 plots recovery against unsupported answering. The edit's trade-off was better than that of the added instruction (B2, via O2, Holm-adjusted $p = 0.005$), activation steering (B4, O2, $p = 0.033$) and directional orthogonalisation (B5, O1, $p = 0.007$). Against B2 and B4 the edit gave fewer unsupported answers without losing recovery; against B5 it recovered more without adding unsupported answers.

Table 5: Sealed confirmation results on H. Rates are base-weighted means (section 3); U is utility with error penalty 1. The last column reports whether the selected edit's trade-off is better than that comparator's, with the orientation that passed and its Holm-adjusted p-value across ten tests. Values were recounted independently from per-row outputs.

	Recovery ↑	Unsupp. ↓	Damage ↓	CF-follow ↑	U ↑	Edit better? (test, p_{Holm})
B0 unedited	0	0.035	0	0.632	0.317	—
Selected edit (rank-1, L35, dose 2)	**0.816**	**0.042**	**0.030**	**0.917**	**0.435**	—
B1 logit bias (−8)	0.793	0.095	0	0.919	0.406	inconclusive
B2 "Do not reply unknown"	0.721	0.083	0.046	0.882	0.389	yes (O2, 0.005)
B3 "unknown" option removed	0.977	0.682	0.021	0.975	0.130	not established
B4 layer-29 steering ($\alpha = 1$)	0.786	0.078	0	0.921	0.418	yes (O2, 0.033)
B5 orthogonalisation ($\beta = 1$)	0.678	0.031	0.099	0.838	0.401	yes (O1, 0.007)

Logit bias: inconclusive. The comparison with B1 is the most informative result in the study, because B1 is the simplest intervention that targets the output directly. The point estimates favour the edit: recovery 0.816 against 0.793, and unsupported answers 0.042 against 0.095. The difference in unsupported answers is clear (edit minus B1: −0.054, 95% CI −0.069 to −0.038). The difference in recovery is not (+0.023, 95% CI −0.043 to +0.088). O2 therefore depends on whether the edit's recovery is at most 0.05 below B1's. That non-inferiority component gave $p = 0.015$, which did not survive Holm correction (adjusted $p = 0.092$). Equivalence was not shown either: TOST gave $p = 0.21$ for recovery and 0.70 for unsupported answering. The data do not show that the edit is better than logit bias, and they do not show that the two are equivalent. B1 also has an advantage the trade-off tests do not score: it broke no previously correct answers.

Option removal: not established. Removing the "unknown" option (B3) recovered almost every false abstention (0.977), but it answered 0.682 of unanswerable questions. The edit recovered less (−0.160) and gave far fewer unsupported answers (−0.641). Neither orientation passes: O1 fails because the edit recovers less, and O2 fails because the recovery loss exceeds the 0.05 margin. The frozen tests cannot rank two arms that sit at such different points of the trade-off. On utility, which weighs both sides equally, B3 scored 0.130 against the edit's 0.435, but utility was not a pre-specified comparison between arms.

False abstentions whose note names the city. In 232 of the 261 false abstentions on H, the unedited model's note names the gold city (by the frozen answer-bearing rule). On this subset, the edit and B1 recovered at similar rates (0.864 and 0.881). This comparison is descriptive and was not tested.

6.3 Transfer to unseen cities

T-city has 24 bases per task and uses cities that appear in no other split. It was read once and is a descriptive check, not a second confirmation. The results resemble those on H. Utility was 0.422 against 0.303 for the unedited model, recovery 0.817, unsupported answering 0.062 against 0.059, damage 0.031 and counterfactual following 0.912. Comparator utilities fell in the same order as on H: B4 0.415, B1 0.412, B5 0.409, B2 0.397, B3 0.107. T-city changes only the answer vocabulary. It does not test new tasks or reworded instructions.

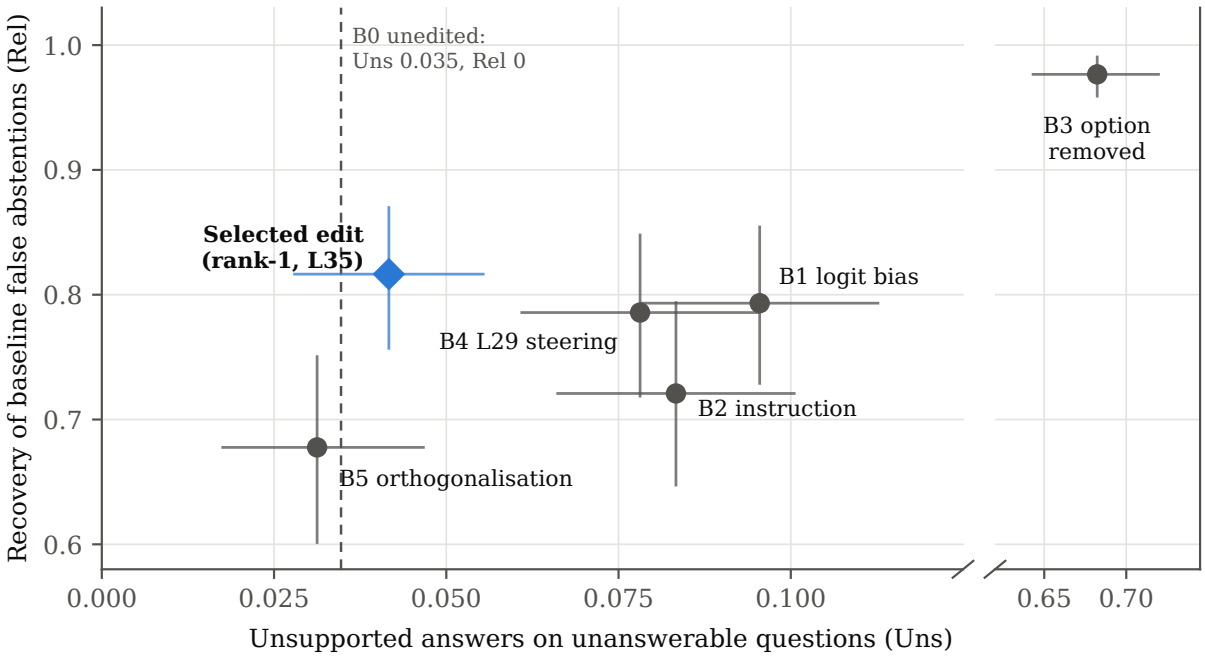


Figure 3: Recovery against unsupported answering on H for the selected edit and the five repair comparators. Up and to the left is better. Bars are marginal 95% bootstrap intervals for each rate, computed after confirmation from the stored per-row outputs (appendix E.3). They are descriptive. The confirmatory comparisons in table 5 use paired differences and Holm correction, so overlapping marginal intervals do not imply that a comparison failed, and separated ones do not imply that it passed. The dashed line marks the unedited model's unsupported-answer rate; its recovery is 0 by definition. The horizontal axis is broken to show B3.

The edit's selection-split results were also close to its confirmation results (utility 0.429 on S and 0.435 on H; counterfactual following 0.910 and 0.917). We report this consistency as description, not as extra evidence, because S was used to select the edit.

7 What the edits explain

An edit can improve behaviour without explaining why the original model failed. We distinguish three levels of evidence, each answering a different question (table 6). The levels are tested on different edits. The repair result concerns the layer-35 rank-1 edit. The pathway and account tests concern a different submission, the two-head edit, because the pathway branch selected it on S. Nothing in this section tests the layer-35 edit's own explanation.

Setting. The pathway tests use the *controlled-prefix* setting. The model is given the unedited model's own note, token for token, and generates only the answer, so differences in the answer cannot come from a changed note. The endpoint is the share of false abstentions answered with the gold city. A shared text prefix does not imply a shared internal state, because an edit changes the activations computed over that prefix.

Table 6: Three levels of evidence and what this study found at each.

Level	Question	Edit tested	Result on H
Behavioural repair	Does the edit improve outcomes on unseen cases?	Rank-1, layer 35	Shown (section 6)
Pathway	Does manipulating the named component change the behaviour?	Two heads, L32h2 and L33h10	Shown at layer 33: sufficient, not necessary
Mechanistic account	Do the observations support the agent's description of what the component computes?	Two heads (account of L33h10)	Not established

7.1 Pathway evidence: a route through layer 33

The two-head edit zeroes the output columns of head 2 in layer 32 and head 10 in layer 33. At its selected dose of 1.5 it is a strong intervention, and it is not selective: on S it produced unsupported answers on 96% of unanswerable rows (total effect). Its role in this study is to locate where an answer-release effect can be produced, not to serve as a repair.

M1: the named component and random edits. Reverting only the layer-33 part of the edit, inside the edited model, reduced recovery by 0.293 (95% CI 0.225 to 0.368). The full edit also recovered 0.717 (95% CI 0.664 to 0.770) more than the mean of five random edits of the same size (B6). Both tests passed (Holm-adjusted $p < 0.001$ across the three pathway tests).

M2: transplant and clamp. Two activation interventions asked whether the layer-33 change can produce the effect alone, and whether the effect needs it. Adding the edit's change in the layer-33 output, at the answer position only, to the *unedited* model recovered 0.808 of false abstentions (95% CI 0.744 to 0.870). Clamping that output back to its original value in the *edited* model removed only 0.193 (95% CI 0.138 to 0.253). The test passed ($p < 0.001$). Under these conditions the layer-33 change is therefore sufficient to release answers but not necessary for the edited model to release them. The remaining release presumably travels by another route. The layer-32 head is a candidate, since reverting it removed a smaller share on S (0.078, against 0.264 for layer 33), but that route was not tested directly.

What the pathway tests do not show. The control B6 applies random Gaussian changes to the same weight entries, matched in size column by column. M1 therefore shows that the agent's specific change to those columns matters more than a random change of the same size. It does not show that L32h2 and L33h10 are unusual among late attention heads; zeroing other pairs might release answers as well. The tests also say nothing about the selected layer-35 repair, which acts at a different layer through a different kind of edit.

7.2 Mechanistic account: not established

The agent described L33h10 as writing an evidence-independent "unknown" prior. It defined the variable f as the projection of the layer-33 attention output, at the answer position, onto that head's mean output direction. The frozen account test M3 had three parts and passed only if all of them did.

(a) *Activity predictions in the unedited model.* The agent predicted how f would change under four manipulations of the memory. All four passed their frozen tests. Removing the evidence lowered f (-1.08, 95% CI -1.54 to -0.62), as predicted. Changing the city (-0.16), adding distractors ($+1.03$, 95% CI 0.52 to 1.55) and replacing absent evidence with unrelated evidence (-0.04) each stayed within frozen equivalence margins of 5.2 to 6.6 units, set at twice the standard deviation on S. The distractor effect nonetheless excludes zero and is larger than the agent's own stated bound of 0.5.

(b) *A separating intervention.* Setting f directly in the unedited model was compared with the logit bias whose effect on S was closest to it. If f carries answerability information, its effect should differ between answerable and unanswerable questions more than the bias's effect does. The interaction was 0.0005 (95% CI -0.025 to 0.027; $p = 0.49$).

(c) *Frozen prediction.* Adding f to a frozen logistic model that predicts which answerable rows end in false abstention, rather than a correct answer, did not improve it on H: the change in AUROC was -4.5×10^{-5} ($p = 0.82$).

Parts (b) and (c) did not pass, so M3 was not rejected (Holm-adjusted $p = 0.82$). Setting f behaved like a generic push away from "unknown", not like an answerability signal.

Why this is not a refutation. Two features of the tests limit what the negative result means. First, the baseline predictor in (c) already had an AUROC of 0.998, so almost no improvement was possible. Second, the tests fit the agent's account poorly. An evidence-independent prior should *not* separate answerable from unanswerable questions, so a null interaction in (b) is close to what the account itself predicts. The interaction test was built to detect an answerability signal, and the agent's account claimed the opposite. The correct reading is that the pre-registered tests did not establish the account. They did not show it to be false.

7.3 Exploratory observations on E

On the exploration split, doubling the two heads (submission 3) turned 180 of 572 correct answers (31%) into "unknown", inside the agent's predicted range of 30–40%. Removing the heads recovered most false abstentions on that split. Together these suggest that the heads act in both directions. The agent built its edits on E, however, so these observations cannot confirm that interpretation (appendix A).

8 Discussion

What the repair establishes. Under a design that kept the agent away from the selection and confirmation data, an agent-proposed weight edit improved this model's handling of false abstention on unseen cases. It met a criterion fixed in advance, and it did better than three of five simpler interventions under corrected tests. The result is an existence proof under controlled conditions. It shows that a closed-form, agent-designed rank-1 edit can move this model to a better point on the recovery–unsupported-answer trade-off than several standard fixes. It does not show that agent-proposed edits generally do so, or that this edit is the best of its kind.

Why the logit-bias comparison matters. A logit bias against "unknown" changes nothing inside the model; it only makes one answer less likely at the output. If an internal edit cannot clearly beat it, the edit's benefit may amount to an output-level shift that happens to be implemented in

the weights. Our data leave this open. The edit gave clearly fewer unsupported answers than B1 at similar recovery, but the test could not rule out a recovery deficit larger than the 0.05 margin after correction, and B1 broke no correct answers while the edit broke 3%. A larger confirmation set, or a comparison targeted at B1 alone, would be needed to resolve it.

Is there an edit-specific mechanism? The evidence does not yet show one. The layer-35 edit was selected and confirmed on behaviour. Its own explanation, an "answer available" signal read by the edit, was not tested in the confirmation; the pathway branch tested the other submission. Whether the edit's read direction matters, as opposed to its write direction toward cities, is the first question of the follow-up below.

Improving behaviour is not the same as explaining it. The study separated the two on purpose, and they came apart. The agent's two-head edit located a component that is sufficient to release answers, yet the account the agent gave of that component was not established. The agent's predictions were often numerically close (31% of correct answers flipped to "unknown" on E, inside its predicted 30–40%), but close predictions on the agent's own data are weak evidence for an account. A useful edit and a plausible story can both be produced without the story being right. Evaluations of agent-proposed interventions should test the two separately.

One model and synthetic tasks. Every confirmatory result concerns one 3B model on generated city questions with a fixed instruction and note format. The false-abstention behaviour itself was much rarer in the three other small models we had examined (section 2.1), so the problem this edit solves may be specific to this model and prompt. The T-city check varies only the answer vocabulary. Natural questions, reworded instructions and misleading contexts [18] remain untested.

The dose grid. The selected dose was the top of its frozen grid, and utility on S was still rising there. The orthogonalisation comparator was also selected at the top of its grid ($\beta = 1$). A wider grid could change the comparison in either direction: a larger edit dose might recover more but also break more correct answers. The edit should be read as one point on a dose curve that was not fully explored.

One agent run. The study observed one session of one agent. Another session, another prompt or another model could propose different edits, better or worse. We did not test whether access to weights was necessary for the agent's success; in a related auditing setting, a calculation tool mattered more than weight access [12]. Estimating how often agents produce edits that pass this evaluation would require many runs.

Controls still needed. Three controls would sharpen the conclusions: random-head and random-direction controls, to test whether the chosen components and directions are special; natural-question data with the evidence in view; and comparisons with component-level methods such as Steer2Edit [19], weight-folded false-refusal ablation [21], gated refusal edits [4] and SAE-feature steering [3]. Our comparisons do not establish superiority over editing or steering methods in general.

8.1 Pre-registered follow-up (in progress; no results reported)

A follow-up was pre-registered after the confirmation closed, to ask whether the two edits are specific. It uses a new controlled split, N (seed 54, 1,950 rows), disjoint from the agent's exploration split and

the selection split, and it cannot change any confirmation verdict above. Its design and manifest were committed before any held-out output was generated (appendix E). **No follow-up results exist at the time of writing, and none are reported here.**

- *Does the layer-35 read direction matter (C4)?* Twenty control edits keep the edit's write vector but use a random read vector. Each is scaled on development rows to match the edit's root-mean-square gate and is sign-aligned to push in the same direction. C4 passes if the edit's utility exceeds all twenty controls and its unsupported-answer rate is below their median. A failure would leave read-side specificity unestablished; on its own it would not show equivalence to an output bias.
- *Are the two heads unusual (C1, C2)?* C1 requires ablating the target pair to recover more false abstentions than each of twenty random pairs from the same layers, by at least 0.30 over their median. C2 requires doubling the target pair to create more abstentions on previously correct rows than doubling each random pair.
- *Does the layer-35 signal follow the agent's predictions (C5)?* Four frozen predictions concern how the proposed signal changes when evidence is removed, the city changes, distractors are added, or unrelated evidence replaces absent evidence.

C1, C2 and C4 require an empirical rank of 1 among 21 evaluated edits; C5 uses an intersection–union test at 0.05. No family-wise error control is claimed. Under a null in which the target's scores are exchangeable with the random controls' (ties counted against the target), and given a valid C5 test, a union bound puts the chance of at least one false pass at about 0.19 or less. This bound holds only under that null; it is not a guarantee against every possible mechanism. Descriptive arms examine single heads, other late-layer pairs, random write vectors, direct logit attribution and the prompt without its "unknown" option. A natural-question check uses 138 LongMemEval development questions with the evidence in view.

9 Limitations, reproducibility and resources

Main limitations.

- *Scope.* One model (Qwen2.5-3B-Instruct), synthetic city questions in three task templates, one instruction and note format, and one agent run. The planned reworded-instruction split was dropped before the first freeze, so transfer to new instructions is untested.
- *Unresolved comparison.* The comparison with logit bias is inconclusive (recovery difference 95% CI -0.043 to $+0.088$); neither superiority nor equivalence was established at the frozen margins.
- *Costs of the repair.* The edit broke previously correct answers (base-weighted damage 0.030, upper bound 0.055), while B1 and B4 broke none. Counterfactual following (0.917) cleared its 0.90 threshold by a small margin. Both thresholds were applied to point estimates.
- *Dose.* The selected dose was the largest in the frozen grid, and larger doses were not searched.
- *Controls.* The only random-edit control in confirmation was B6, five size-matched perturbations of the same weight entries, used in the pathway tests. Random-head and random-direction controls belong to the follow-up. The prompt offers "unknown", and the present results do not distinguish a change in how the model treats that option from a change in how it uses the evidence.

- *Mechanism tests.* The account test faced a predictor ceiling (baseline AUROC 0.998) and fitted the agent's account poorly (section 7). The pathway result is sufficiency, not necessity.
- *The note.* The note is a prompted summary. We do not treat it as a faithful trace of the model's reasoning.

Reproducibility. The design was frozen in three stages before the data each stage governed were generated: F0 (data, grader, comparators, edit rules), F1 (bounds, run procedure, cost limits) and F2 (candidates, doses, analysis code). All model outputs on H and T-city were generated after F2. Besides two deviations recorded at F1, four tooling fixes were made: a comparator file missing from the run bundles, smoke tests that could not apply full-size edits, a missing key that crashed the first selection run ($0.09, counted in the cost below), and a guard failure that exited without a log message. None changed a scoring, selection or decision rule, and each was reviewed and accepted in the project record. Appendix E lists the commits, hashes, code and per-row outputs, and states which artefacts are not yet public.

Compute and cost. The agent run cost $3.58 in API usage. Replaying the submissions on E took about 59 hours on a laptop CPU at no cloud cost. Selection on S ran on one NVIDIA A40 GPU and cost $4.38, including $0.09 for a failed first attempt; confirmation on H and T-city took 4.36 hours on one A40 and cost $2.57. Total GPU charges were $6.95.

10 Conclusion

This case study asked whether an AI agent can propose a weight edit that repairs a specific failure, Qwen2.5-3B answering "unknown" with the answer in its own note, without teaching the model to answer what it cannot support. Under an evaluation that kept the agent's data, the selection data and the confirmation data apart, the answer for this model and task is yes. A rank-1 edit to the last attention layer recovered most false abstentions on sealed data, added few unsupported answers, and beat three of five simpler fixes on pre-specified trade-off tests. It broke a small share of correct answers, and it did not clearly beat a logit bias against "unknown".

The study does not show that the agent understood why the model fails. The component it named for a different edit lies on a route that can release answers, but the agent's account of that component was not established, and the selected repair's own explanation was not tested. The natural next question is whether the edit is specific: whether its read direction and the agent's chosen heads matter, and whether its advantage over output-level fixes survives natural questions and other models. The pre-registered follow-up in section 8.1 addresses the first part of that question.

Disclosure of AI assistance

The weight edits studied here were proposed by Claude Opus 5.5 (Anthropic), run through the API as the agent described in section 4. AI assistants (Claude and GPT models) also helped design the study, write and review the analysis code, independently recount the results from per-row outputs, and draft and edit this report. The author reviewed the design, the results and the text, and is responsible for the content. [TODO: Author to confirm this wording, in particular the final sentence.]

Unresolved before publication (draft only)

1. **Public artefacts.** No public repository URL or archive DOI yet; the pre-registration order can only be checked by readers once the commit history is public (appendix E).
2. **Edited weights and transcript.** Confirm the model licence permits releasing edited weights; confirm the redacted agent transcript is ready to publish.
3. **Disclosure wording.** Author to approve the AI-assistance statement.
4. **Follow-up.** Section 8.1 has no results. Either keep it as a labelled in-progress section or remove it before release; do not add results until the follow-up is recounted and reviewed.
5. **Bibliography caveats.** Kumaran et al.: journal version not confirmed from a primary source; commit–abstain circuit: NeurIPS 2026 venue taken from the arXiv comment; GRAB-RAG: AACL-IJCNLP 2026 acceptance not confirmed.
6. **Activity-prediction count.** This draft reports that all four M3 activity tests passed, from the sealed report; v0.2 and the H review said "two". Author to confirm (see the editorial review file).
7. **Informal links.** The two project write-ups cited in section 2.1 are informal web pages; decide whether to keep them.

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A Exploratory results on the agent's split (E)

These numbers come from E, the split the agent studied while designing its edits. They determined which submissions entered the evaluation. They are not evidence that the edits generalise. E has 144 base cases (48 per task) and 1,872 rows; the unedited model abstains falsely on 284 answerable rows, spread over 94 bases (DE 44, RC 35, TU 15). All edits were replayed at dose 1 in the total-effect setting unless stated.

Table 7: Replay of the three submissions on E (dose 1). Rates are base-weighted. Entry intervals are one-sided 97.5% bootstrap bounds. The pathway effect is false abstention → gold for edits 1 and 2 and correct → abstention for edit 3, measured in the controlled-prefix setting.

	B0	Edit 1 (zero heads)	Edit 2 (rank-1 L35)	Edit 3 (double heads)
Recovery	0.000	0.862	0.739	0.009
Unsupported	0.040	0.076	0.050	0.030
Damage	0.000	0.028	0.015	0.381
CF-follow	0.653	0.940	0.887	0.451
Degenerate	0.005	0.006	0.003	0.013
Utility ($\lambda = 1$)	0.305	0.427	0.427	0.198
Repair entry: $U - U_{\mathrm{B0}}$	—	+0.122	+0.122	−0.107
bounds		(+0.102, +0.141)	(+0.104, +0.139)	(−0.126, −0.088)
Pathway entry: signed effect	—	+0.817	+0.706	+0.360
bounds		(+0.753, +0.877)	(+0.633, +0.777)	(+0.308, +0.410)
Entered	—	repair, pathway	repair, pathway	pathway only

Row counts. Edit 1 released 252 of 284 false abstentions and kept 561 of 572 correct answers. Unsupported answers rose from 6 to 15 of 432 related-distractor rows and from 17 to 29 of 144 swapped-name rows. Edit 2 released 222 of 284 and kept 563 of 572; unsupported answers rose from 6 to 11 and from 17 to 18. Edit 3 turned 180 of 572 correct answers into "unknown" (the agent had predicted 30–40%).

Note-then-refuse on E. By the frozen answer-bearing rule, the note names the gold city in 267 of the 284 false abstentions on E (taken from the stored population labels of the unedited model).

B Claims and evidence

Table 8: Every substantive claim in the report, the evidence for it and its status. "Shown" requires a passed pre-specified test on H; "context" marks results of earlier experiments.

Claim	Evidence	Status
Earlier experiments		
False abstentions in this model are mostly "note, then refuse"	152 of 165 held-out false abstentions name the gold in the note (earlier experiment); 232 of 261 on H (this study)	Context; descriptive here
Ungated pushes recover answers but increase unsupported answering	Earlier held-out results (0.188 and 0.451); a gating result was template-limited	Context
This study: repair branch		
At least one agent-proposed edit entered the evaluation	Entry on E (appendix A)	Exploratory
The rank-1 layer-35 edit repairs false abstention	H, four-part repair criterion (section 6)	Shown
Its trade-off is better than B2, B4 and B5	H, Holm-corrected O1/O2	Shown
Its trade-off is better than B1	H, O1/O2 not passed; TOST not passed	Inconclusive
Its trade-off is better than B3	H, O1/O2 not passed	Not established
It carries over to unseen cities	T-city	Descriptive
This study: pathway branch (two-head edit)		
The release runs through the layer-33 component	H, M1 and M2	Shown: sufficient under the tested intervention, not necessary
The two heads are unusual among late-layer heads	No random-head control in confirmation	Not tested
The agent's account of what the component computes	H, M3	Not established (not refuted)
Follow-up in progress (section 8.1); no results		
The layer-35 edit's read direction matters (C4)	N split	Pending
The two heads are special among late-layer pairs (C1, C2)	N split	Pending
The agent's description of the layer-35 signal (C5)	N split	Pending
The edits carry over to natural questions	LongMemEval development questions	Pending (descriptive)

C Methods in detail

C.1 Prompt

The user turn for case `E-DE01_ANSa_H0` (fig. 1) is reproduced below. It is wrapped in the model's default chat template, whose system turn reads "You are Qwen, created by Alibaba Cloud. You are a helpful assistant." Only the memory lines and the question vary between cases.

```
Below is your memory of a long conversation, followed by a question about it.  Answer with a
short phrase that states the fact asked for.  If the memory does not contain the answer,
reply:  unknown
Each memory line shows who said it and when.  First write a brief note of at most 80 words:
name the facts that bear on the question in a few words each (do not copy whole lines), then
do any counting, adding or date arithmetic.  Give dates as absolute dates (for example 7 May
2023), never relative ones such as 'yesterday' or 'last year'.  If the question asks for
advice or a suggestion, answer with a brief suggestion that uses what the memory says about
the user.  Reply unknown only if the memory has nothing at all on the topic.  End with one
line of the form 'Answer:  <short phrase>'.

## Memory
user, 2023/03/01 (Wed) 10:00:  Ben lives in Manchester.
user, 2023/03/02 (Thu) 10:00:  Ben works as a manager.

## Current input
Question:  (asked on 2023/03/03 (Fri) 12:00) Where does Ben live?

Note:
```

B2 appends "Do not reply unknown." as a new line after the final instruction sentence. B3 deletes the two sentences that mention "unknown": "If the memory does not contain the answer, reply: unknown" and "Reply unknown only if the memory has nothing at all on the topic."

C.2 Answer span and grading

The grader reads the text after the last `Answer:` marker and scores the preceding note separately. A missing marker is its own class and counts toward the degenerate-output rate. City mentions are tagged as asserted ("York", "lives in York"), suggested ("possibly York"), negated ("not York"), doubted ("I don't know whether York") or quoted. Only asserted and suggested mentions count as answers. If there is no asserted or suggested city, an abstention phrase gives the abstention class. One city is graded as gold, wrong or other content; one city together with an abstention phrase is mixed; two or more distinct cities are conflicting. The grader and its test fixtures were frozen before the agent saw E. Whether a note is *answer-bearing* (names the gold city) is decided by a separate frozen rule applied to the note text.

C.3 Utility

Row type	Gold	Abstention	Mixed	Wrong, conflicting, other, degenerate or missing
Answerable, counterfactual	$+1$	0	0	$-\lambda$
Unanswerable	—	0	$-\lambda$	$-\lambda$

$\lambda = 1$ is the primary setting and $\lambda = 2$ a secondary one. Each base contributes equally to aggregate utility, and within a base the answerable and unanswerable rows each supply half of the weight. Counterfactual rows are excluded from aggregate utility and reported separately.

C.4 Two intervention settings

In the *total-effect* setting, the model generates both its note and its answer under the intervention. This is the primary setting for the repair branch. In the *controlled-prefix* setting, the intervened model is fed the unedited model's stored token IDs up to and including the answer marker and then generates only the answer (at most 16 tokens). This is the primary setting for the pathway branch. The shared text prefix does not imply a shared internal state.

C.5 Applying edits

Each edit is stored as a hashed manifest of components. A dense component is stored as factors U and V, and its change is $\Delta W = (s\,U)V^\top$ for dose s; dose 1 is the edit as saved. Sparse components store the changed rows or columns. Edits are added to the fp32 weights, and each result is checked against a hash recorded in the manifest. B6 random edits draw Gaussian changes on the same entries as the target edit, matched in Frobenius norm column by column, with seeds 1 to 5.

C.6 Inference

The paired percentile bootstrap resamples base cases within each task, with $B = 10{,}000$ attempted draws and a fixed seed. Tests use approximate percentile inversion. For a one-sided null $\theta \le \theta_0$, the p-value is

$$p = \frac{1 + \#\{b : \theta_b^* \le \theta_0\}}{B_{\text{valid}} + 1}.$$

Undefined replicates, such as those with an empty denominator or an undefined AUROC, are excluded and counted. If more than 1% of attempted draws fail, the statistic receives no decision. If all valid draws equal the point estimate, the frozen rule assigns $1/(B_{\text{valid}}+1)$ when that estimate is on the alternative side of the threshold, and 1 otherwise. These are approximate bootstrap tests, not exact randomisation tests.

Equivalence uses two one-sided tests (TOST) with margin ± 0.05. For an intersection–union claim, the composite p is the largest elementary p. The repair family contains O1 and O2 against B1–B5 (ten hypotheses), and the pathway family contains M1–M3. Each family is Holm-corrected at 0.05; no study-wide guarantee is claimed. Marginal 95% intervals describe uncertainty, and the corrected tests determine the confirmatory decisions. The 0.05 margins, the 0.90 counterfactual-following threshold and the 20-base minimum are operational conventions fixed before the data they govern were generated.

C.7 Stopping rules

H and T-city were each generated and read once, with no rescue variant and no second agent run. If no candidate had met the screening rule on E or the entry rule on S, the study would have stopped and documented that outcome without opening H or T-city.

D Related work in detail

Table 9: The closest work and how this study differs. Descriptions follow each paper's abstract or main text.

Work	What it did	How this study differs
Agents and weight edits		
Weight-space programming [20]	An LLM agent improved Qwen3.5-0.8B's maths by removing a reasoning-loop direction, changing 0.016% of its weights; checked on held-out benchmarks	Target behaviour has no wanted refusals to protect; here unsupported answers on unanswerable questions are penalised, and confirmation is sealed and pre-specified
Steer2Edit [19]	Converts steering vectors into edits of individual heads and neurons	Not agent-proposed; not among our frozen comparators
Interpretability agents [2, 17]	Agents that describe components, or a proposed agenda for agents that choose and apply interventions	Here one agent-proposed intervention is tested against pre-specified bars
LLMs as auditors [12]	Single-prompt LLM audits (and repairs) of small clinical networks; a calculation tool mattered more than weight access	Language models and agentic editing; a caution that an agent's account of the weights may not drive its success
Knowledge editing [13, 14]	Locate and edit stored factual associations	Our target is a decision behaviour, not a stored fact
Refusal, abstention and their components		
Refusal and unanswerability directions [1, 8, 9]	A single direction mediates refusal; a linear direction tracks unanswerability; conditional steering makes interventions input-dependent	Our earlier ungated pushes increased unsupported answering; an earlier gate was template-limited
Safety heads [23]	Ablating single attention heads in Llama-2-7b-chat sharply raises compliance with harmful requests	There the refusal is wanted; ours is an unwanted "unknown" with the evidence in view
Retrieval heads [22]	Under 5% of heads perform in-context retrieval; pruning them causes hallucination, pruning random heads does not	Our interventions target false abstention with sufficient evidence; the role of the selected heads is unresolved, and random-head controls are part of the follow-up
Commit–abstain circuit [16]	Describes a commit-versus-abstain circuit across ten models (3B–14B); a policy trained on its activations reduces false abstentions	The closest target behaviour; their fix is a trained activation-level policy, not an agent-proposed weight edit, and is not pre-registered
Abstention as prompt artefact; answer release [10, 11]	Abstention can follow the "unknown" option; suppressed answers can often be released, and release is more local than refusal restoration	Predicts that a late edit can release answers; that part of our result is expected
Abstention behaviour [5–7]	Models misuse sufficient context; models fail to abstain on unanswerable questions; internal confidence drives abstention	Behavioural or confidence-based accounts; no component-level edit for in-view false abstention

Continued on next page

Work	What it did	How this study differs
Selective interventions		
False-refusal vector ablation [21]	Removes a false-refusal direction from the weights; scores released false refusals against kept true refusals	Safety over-refusal, not evidence in view; not agent-proposed; not among our comparators
Residual Paving [4]	Trained, gated residual modules on a frozen backbone edit safety refusals, scored on three objectives on a held-out split	The closest selectivity yardstick; safety refusal, trained modules, not agent-proposed, not pre-registered
Dr. No [3]	SAE-feature steering that trades answering against abstention, mainly in vision-language models	Activation steering only; no weight edit, no pre-registration
Evaluation practice and benchmarks		
Pre-registered interpretability [15]	Pre-registered, once-only held-out tests of representation steering in learned fluid models	Precedent for the method, outside language models
GRAB-RAG [18]	Small frozen RAG models under supportive, missing, degraded and misleading context; prompt-based abstention policies and verifiers	No internal (weight or activation) intervention; a source of natural counterfactual evidence for future tests

E Artefacts and reproducibility

E.1 Artefacts

Commit identifiers refer to the project's Git repository, which is not yet public. [TODO: public repository URL; archive DOI if used.]

- *Pre-registration and freezes.* Proposal revisions `d9e67b7`, `1be9281`, `15ec3ee`; F0 `d29263a`; F1 `1a37a72` (built at `c9ca458`); F2 `02bb2f6` (design hash `d02d9f9c...`).
- *Tooling fixes.* `b6e3d8e` (before any GPU run), `4bf49ad` (after the failed first selection run), `c880239` and `bf22ac3` (before the confirmation run).
- *Confirmation report.* `h_report.json`, sha256 `51bbbd2e...`, committed at `321a4f1`.
- *Cases.* 5,928 rows in 456 base cases, sha256 `5eeb7c40...`.
- *Edit recipes.* Submission 1 `c64ea090781450cf`, submission 2 `8364bf135ac59a83`, submission 3 `87224655b7a5361e` (head columns and rank-1 factors). Edited weights will be released only if the model licence allows.
- *Agent record.* The full transcript (paths and keys removed) and the three submission records.
- *Per-row outputs* for every arm and split, and the scripts for fig. 3 (`figures/tradeoff_data.py`, `figures/tradeoff_plot.py`).
- *Follow-up (section 8.1).* Proposal revision 1b `ea88023`; build `9ff3d61`; sign rule `1dce499`; manifest sha256 `b2b14bac...`, committed at `a177536` before any held-out output; N split of 1,950 rows, sha256 `651c930c...`. [PENDING follow-up output commit and report hash.]

E.2 Reproducibility checklist

Table 10: Reproducibility checklist.

Item	Where it is specified	Status
Model and revision	Qwen2.5-3B-Instruct, Hugging Face revision `aa8e7253...`, recorded at F0; fp32	Specified
Prompt and decoding	Appendix C.1; greedy; 160 new tokens (16 for answer-only)	Specified
Case generation and splits	Seeds 51–53; base-level partition (section 3); case-file hash	Specified; generator code not yet public
Grading rules	Appendix C; frozen grader with fixtures	Specified; code not yet public
Edit recipes	Hashed manifests; section 4 gives the selected edit's form	Specified; weights pending licence check
Selection rules	Section 5	Specified
Frozen analysis	F2 commit; inference rules in appendix C	Specified; code not yet public
Agent provenance	Transcript, submission records, API usage log	Held; redacted release pending
Per-row outputs	All arms and splits	Held; not yet public
Compute and cost	Section 9	Reported

E.3 Data for the trade-off figure

Table 11 lists the values plotted in fig. 3. The intervals were computed after confirmation from the stored per-row outputs on H, with the frozen bootstrap (base resampling within task, $B = 10{,}000$, frozen seed). No model was rerun. The script checks that each point estimate matches the sealed report exactly. These marginal intervals are descriptive and were not part of the frozen analysis.

Table 11: Rates on H with marginal 95% bootstrap intervals. Recovery counts are gold answers among the 261 false abstentions; unsupported counts are among the 576 unanswerable rows (every base has four, so the base-weighted rate equals the pooled fraction).

Arm	Recovery [95% CI]	Rows	Unsupported [95% CI]	Rows
B0 unedited	0	0/261	0.035 [0.023, 0.047]	20/576
Selected edit	0.816 [0.756, 0.871]	212/261	0.042 [0.028, 0.056]	24/576
B1 logit bias	0.793 [0.728, 0.855]	210/261	0.095 [0.078, 0.113]	55/576
B2 instruction	0.721 [0.646, 0.795]	190/261	0.083 [0.066, 0.101]	48/576
B3 option removed	0.977 [0.958, 0.992]	252/261	0.682 [0.642, 0.720]	393/576
B4 steering	0.786 [0.718, 0.849]	209/261	0.078 [0.061, 0.095]	45/576
B5 orthogonalisation	0.678 [0.600, 0.751]	184/261	0.031 [0.017, 0.047]	18/576